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Assignment No 2

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✂ SCENARIO Imagine you are a university student. Your exam is 7 days away. You have: Several topics to prepare
Some difficult topics Limited study time Need for daily revision Need for practice questions Your goal is to use AI to create a practical and realistic 7-Day Study Plan.

□ PART 1 — TEST THE BASIC PROMPT

acha maree bat suno mara 7 days bad c++ oop ka mara university ka paper hay to ois kay leya mujay tum guide karho kasay tayari karho

Bilkul Jan 🙌 Agar 7 din baad C++ OOP ka university paper hai, to hum random study nahi karenge. Hum 7-day exam-focused preparation plan banayenge jisme theory + coding + past-paper style questions dono cover honge.

🎯 Hamara Goal

7 din mein:

- C++ OOP ke important concepts clear karna
- Har concept ke easy examples/code karna
- Important theory questions prepare karna
- Daily practice questions solve karna
- Last 2 days mein revision + mock paper karna

📅 7-Day C++ OOP Plan

Day	Main Topics	Practice
Day 1	C++ basics + OOP introduction + Classes & Objects	5 basic programs
Day 2	Constructors, Destructors, <code>this</code> pointer	5 programs
Day 3	Encapsulation + Access Specifiers + Getters/Setters	5 programs
Day 4	Inheritance + Types of Inheritance + <code>super</code> equivalent concepts	6 programs
Day 5	Polymorphism + Function Overloading + Function Overriding + Virtual Functions	6 programs
Day 6	Abstraction + Abstract Classes + Friend Function + Static Members + revision	Important questions
Day 7	🔥 Full Revision + Mock Paper	Exam simulation

🧠 Har din ka method

Aapko 3 hours/day mil jayein to:

1st hour — Concept

Main tumhein topic bilkul easy Urdu/English mein samjhaunga.

2nd hour — Coding

Hum examples khud likhenge aur samjhenge ke har line kya kar rahi hai.

3rd hour — Exam Practice

Main tumhein university-style questions dunga:

- MCQs
- Short questions
- Definitions
- Difference questions
- Coding questions
- Output questions

🔥 Sabse important C++ OOP topics

Inko bilkul strong karna hai:

1. Class & Object
2. Constructor
3. Destructor
4. Encapsulation
5. Public / Private / Protected
6. Inheritance
7. Types of Inheritance
8. Function Overloading
9. Function Overriding
10. Polymorphism
11. Virtual Function
12. Abstraction
13. Abstract Class
14. Friend Function
15. Static Members
16. `this` pointer
17. Access Specifiers

Ek important baat: agar tumhare teacher ne syllabus/past papers diye hain, to unko priority denge.
University paper mein aksar teacher ke covered topic ↓ questions aate hain.

17. Access Specifiers

Ek important baat: agar tumhare teacher ne syllabus/past papers diye hain, to unko priority denge. University paper mein aksar teacher ke covered topics se questions aate hain.

🚀 Main tumhare saath 7 din ka preparation kar sakta hoon

Hum Day 1 se proper class ki tarah start karte hain:

Concept → Easy Example → Code → Explanation → Practice → Quiz

Agar tumhare paas C++ OOP ka syllabus, notes, PDF, ya past paper hai, yahan upload kar do. Main usko dekh kar tumhare liye exact 7-day exam preparation plan bana dunga aur phir Day 1 ki tayari start karwa dunga.

📄 🗣️ 📤 🔄 ...

PART 2 — CREATE YOUR IMPROVED PROMPT

Act as my professional **C++ Object-Oriented Programming (OOP)** university exam tutor. My exam is in **7 days**, so create a focused and realistic 7-day preparation plan. Teach me from basic to advanced topics in simple English and Roman Urdu, with special focus on important exam concepts, definitions, syntax, programming questions, MCQs, short questions, long questions, and commonly asked topics. After each topic, give me a few practice questions and test me without immediately showing the answers. Track my weak areas, revise them regularly, and prioritize high-mark topics. Keep everything **simple, practical, exam-focused, and time-efficient**. Start by asking for my syllabus, notes, or past papers if available, then begin my **Day 1** preparation.

Show less ^

Upgrade

Bilkul. Aaj se hum 7-day C++ OOP Exam Bootcamp start karte hain. 🔥

Main tumhara tutor hoon, aur approach ye hogi:

Concept → Simple English → Roman Urdu → Syntax → Example → Exam Questions → Practice → Quiz → Weak Areas → Revision

Pehle mujhe ye material bhejo

Agar available hai to C++ OOP ka syllabus, teacher ke notes/PDFs, assignments, quizzes, ya past papers upload kar do.

Priority:

1.  Official syllabus
2.  Teacher notes/PDF
3.  Past papers
4.  Previous quizzes/assignments

Agar material nahi hai, **ko problem nahi** — hum standard university-level C++ OOP syllabus se start karenge.



7-Day C++ OOP Exam Plan

Day	Topics	Exam Focus	
Day 1	C++ Basics + OOP Introduction + Class & Object	★★★★★	
Day 2	Constructors + Destructors + <code>this</code> Pointer	★★★★★	
Day 3	Encapsulation + Access Specifiers + Getters/Setters	★★★★★	
Day 4	Inheritance + Types of Inheritance	★★★★★	
Day 5	Polymorphism + Overloading + Overriding + Virtual Functions	★★★★★	
Day 6	Abstraction + Abstract Classes + Friend/Static Members + Full Revision	★★★★★	
Day 7	Complete Revision + Important Questions + Mock Paper	🔥🔥🔥	



DAY 1 — C++ OOP FOUNDATION

1. What is OOP?

Simple English

OOP = Object-Oriented Programming

It is a programming approach where we organize our program around **objects and classes**.

Roman Urdu

Simple words mein:

OOP mein hum real-world cheezon ko programming mein **objects** ki form mein represent karte hain.

For example, university mein:

- Student
- Teacher
- Course
- Book
- Bank Account

Ye sab real-world objects hain.



Agar hum Student ko C++ mein represent karein:

```
</> C++  
  
class Student {  
    string name;  
    int age;  
};
```

Yahan `Student` ek class hai.

2. What is a Class?

Definition ★

A class is a user-defined data type that contains data members and member functions.

Roman Urdu:



Roman Urdu:

Class ek blueprint/template hoti hai jiske through hum objects create karte hain.

Real-life example

Socho house ka map.

Map = Class

Actual house = Object

Similarly:

```
Class → Student  
Objects → Ali, Ahmed, Sara
```

3. What is an Object?

Definition ★

An object is an instance of a class.



Example:

```
</> C++  
  
class Student {  
public:  
    string name;  
    int age;  
};  
  
int main() {  
  
    Student s1;  
    Student s2;  
  
}
```

Here:

```
</> C++
```

```
Student
```



is the class.

And:

```
</> C++
```

```
s1
```

```
s2
```



are objects.

Remember this for exam:

```
Class = Blueprint  
Object = Actual thing created from blueprint
```



4. Basic Class Syntax

Ye syntax exam ke liye bohat important hai:

```
</> C++
```

```
class ClassName {  
  
    public:  
  
        // data members  
        // member functions  
  
};
```



Notice:

 C++

};

Class ke end par semicolon ; lagta hai.

5. First C++ OOP Program

 C++

```
#include <iostream>
using namespace std;

class Student {

public:

    string name;
    int age;

    void display() {
        cout << "Name: " << name << endl;
        cout << "Age: " << age << endl;
    }
};
```

</> C++



⚡ Upgrade

```
int main() {  
  
    Student s1;  
  
    s1.name = "Sayyam";  
    s1.age = 20;  
  
    s1.display();  
  
    return 0;  
}
```

Output

Name: Sayyam
Age: 20



Code ko samjho

</> C++



```
class Student
```

Student naam ki class banayi.



</> C++



⚡ Upgrade

```
string name;  
int age;
```

Ye **data members** hain.

</> C++



```
void display()
```

Ye **member function** hai.

</> C++



```
Student s1;
```

Ye **Student** class ka object create karta hai.

</> C++



```
s1.name = "Sayyam";
```

Object ke through data access kiya.

⌞ C++



✦ Upgrade

```
s1.display();
```

Object ke through function call kiya.

★ 6. Data Members vs Member Functions

Ye short question aa sakta hai.

Data Members

Class ke andar variables.

⌞ C++



```
string name;  
int age;
```

Member Functions

Class ke andar functions.

Member Functions

✦ Upgrade

⬆ Share

Class ke andar functions.

⌞ C++



```
void display() {  
    cout << name;  
}
```

Easy trick

Data = information

Function = action

Student:

```
Data:  
name  
age  
rollNo
```



```
Actions:  
display()  
study()  
calculateMarks()
```



7. Access Specifiers

C++ OOP ka very important topic.

Three major access specifiers:

```
public
private
protected
```



public

Public members class ke bahar bhi access ho sakte hain.

```
</> C++
class Student {
    public:
        string name;
};
```



Then:

Then:

```
</> C++
Student s1;
s1.name = "Sayyam";
```



Allowed 

private

Private members directly class ke bahar access nahi kiye ja sakte.

```
</> C++
class Student {
    private:
        int age;
};
```



This is not allowed:



This is not allowed:

Upgrade

```
</> C++  
Student s1;  
s1.age = 20; // Error
```



Roman Urdu

Private ka matlab:

Sirf class ke andar directly access.

protected

Protected members:

Class ke andar aur derived/inherited classes mein accessible hote hain.

Inheritance ke time iska importance zyada samajh aayega.



DAY 1 — Exam Definitions



DAY 1 — Exam Definitions

In definitions ko prepare karo:

Q1. What is OOP?

Answer:

OOP is a programming paradigm that organizes programs around objects and classes.

Q2. What is a class?

A class is a user-defined data type that contains data members and member functions.

Q3. What is an object?

An object is an instance of a class.

Q4. What is a data member?

A variable declared inside a class is called a data member.

Q5. What is a member function?

A function declared inside a class is called a member function.



A function declared inside a class is called a member function.

Q6. Name the three access specifiers in C++.

`public`, `private`, and `protected`.

DAY 1 MCQs

Answers abhi mat dekho/search karo. Tum mujhe sirf answers bhejna.

MCQ 1

OOP stands for:

- A. Object Operating Program
- B. Object-Oriented Programming
- C. Object Order Programming
- D. Organized Object Programming

MCQ 2

An object is:

- A. A function
- B. A variable only



C. A header file

MCQ 3

Which keyword is used to create a class?

- A. `object`
- B. `Class`
- C. `class`
- D. `structclass`

MCQ 4

Which access specifier allows access from outside the class?

- A. `private`
- B. `protected`
- C. `public`
- D. `internal`

MCQ 5

Which one is a member function?

- A.

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A.

Upgrade

```
</> C++  
int age;
```



B.

```
</> C++  
string name;
```



C.

```
</> C++  
void display() {}
```



D.

```
</> C++  
Student s1;
```



DAY 1 Coding Test

Upgrade

Question 1:

Create a C++ class named `Car` with:

- `brand`
- `model`
- `year`
- `display()` function

Then create an object and display the information.

Question 2:

Create a class named `Student` with:

```
name  
rollNo  
marks
```



Create an object and display all three values.



Short Questions

Answer these **without looking back** if possible:

1. What is OOP?
2. What is a class?
3. What is an object?
4. Difference between class and object?
5. What are data members?
6. What are member functions?
7. Name the three access specifiers.
8. What is the purpose of `public` ?
9. What is the purpose of `private` ?
10. Why do we use classes in OOP?

Your Day 1 Task

Abhi tum sirf ye 3 cheezen karo:

1. Agar syllabus/notes/past papers hain → **upload karo**.
2. Upar ke 5 MCQs ke answers bhejo, e.g.

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1-B
2-C
3-C
4-C
5-C



3. Coding Question 1 ka code khud likhne ki koshish karo.

Main tumhare answers check karunga, **weak areas track karunga**, phir Day 1 ka next part karwaunga: **constructors + deeper class/object practice**, aur gradually Day 2 par move karenge.



Do you like this personality?



PART 2 — CREATE YOUR IMPROVED